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2027 CYCLE

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CUET UG

Mathematics

Linear Programming

Formulation, Feasible Regions and the Corner Point Theorem · Section A 6 + B1 V + B2 VIII · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Introduction and terminology	"Introduction, related terminology such as constraints, objective function, optimization" (Section B1 Mathematics Unit V) — "Introduction and related terminology. Familiarize with terms related to Linear Programming Problem." (Section B2 Applied Mathematics Unit VIII)	Naming decision variables, objective and constraints from a worded story.
Mathematical formulation	"Mathematical formulation of Linear Programming Problem. Formulate Linear Programming Problem." (Section B2 Applied Mathematics Unit VIII)	Story → inequalities with non-negativity attached; objective written explicitly.
Graphical method in two variables	"Graphical method of solution for problems in two variables" (Section A Unit 6, compulsory) — "Draw the Graph for a system of linear inequalities involving two variables and to find its solution graphically." (Section B2 Applied Mathematics Unit VIII)	Half-plane shading, region overlap, corner listing, Z evaluation.
Feasible and infeasible regions; bounded and unbounded	"Feasible and infeasible regions (bounded or unbounded), feasible and infeasible solutions" (Section B1 Mathematics Unit V) — "Identify feasible, infeasible and bounded regions." (Section B2 Applied Mathematics Unit VIII)	Reading a region's shape and existence before touching the objective.
Optimal feasible solutions	"Optimal feasible solution" (Section A Unit 6; Section B1 Unit V adds "up to three non-trivial constraints") — "Understand feasible and infeasible solutions. Find optimal feasible solution." (Section B2 Applied Mathematics Unit VIII)	Corner point theorem in action; quoting the optimum point and value correctly.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every

correct answer and −1 for every incorrect answer. This is the syllabus's one triple-appearance unit — **Section A Unit 6 (compulsory)**, **Section B1 Unit V** and **Section B2 Unit VIII** — so the graphical method below is examinable for every candidate in every section, with the syllabus capping the algebra at three non-trivial constraints.

What a Linear Programming Problem Is

Linear programming answers one business-shaped question: **given limited resources, what is the best plan?** The ingredients are always the same three. **Decision variables** x and y — the quantities you control, like units of two products. An **objective function** $Z = ax + by$ — the scoreboard you optimise (profit to maximise, cost to minimise). **Constraints** — linear inequalities carved out of the story's limits (machine hours, raw material, budget), plus the ever-present **non-negativity** conditions $x \geq 0$, $y \geq 0$, because negative production is not a plan. The syllabus caps the algebra at "up to three non-trivial constraints", so every exam problem fits on one clean sketch.

Formulation is itself a scored skill. The translation dictionary: "at most / not more than / within" gives $\leq$; "at least / no less than / minimum requirement" gives $\geq$; "profit per unit" and "cost per unit" become the coefficients of Z. One worked formulation:

WORKED EXAMPLE — FROM STORY TO INEQUALITIES

A carpenter makes tables and chairs. Each table needs 4 hours of carpentry and 2 hours of finishing; each chair needs 3 hours and 1 hour. Available: 240 hours of carpentry, 100 hours of finishing. Profit: ₹70 per table, ₹50 per chair. Formulate the LPP to maximise profit.

Solution. Decision variables: x = tables, y = chairs. Objective:

$$\text{maximise } Z = 70x + 50y.$$

Constraints (carpentry, finishing, non-negativity):

$$4x + 3y \leq 240, \quad 2x + y \leq 100, \quad x \geq 0, \quad y \geq 0.$$

Every sentence of the story has become one line of algebra — nothing invented, nothing lost. The next sections solve exactly this shape.

Half-Planes and the Feasible Region

Each linear inequality owns a **half-plane**: the line $ax + by = c$ splits the sheet, and the inequality claims one side of it. To find which side, use the **test point** trick — the origin is favourite: plug $(0, 0)$ into the inequality; if it comes out true, the origin's side is the claimed side. The **feasible region** is where all the claimed half-planes overlap together with the first quadrant (non-negativity keeps everything right of the y-axis and above the x-axis). A point inside is a **feasible solution** — a plan that obeys every constraint; a point outside is **infeasible** — it breaks some resource limit, no matter how attractive Z looks there.

- **Draw lines first, shade second.** Replace each $\leq$ or $\geq$ by $=$ and plot the straight line from its two intercepts — the boundary is what you actually draw.

- **Strict vs inclusive.** $\leq$ and $\geq$ include the line itself (solid); a strict $<$ or $>$ would exclude it (dashed) — the syllabus's problems are inclusive throughout.
- **The region's mood matters.** If the overlap is a closed polygon, it is **bounded**; if it runs off the sheet, **unbounded**; if the half-planes never all agree, the region is **empty** and the LPP is **infeasible** — a legitimate answer the MCQs test.

The Corner Point Theorem — Where Optimums Live

Here is the theorem the entire method stands on: **if an optimal value of Z exists, it occurs at a corner (vertex) of the feasible region**. The intuition is a sliding ruler: $Z = ax + by$ is a family of parallel "iso-profit" lines; as you slide the ruler to improve Z, the last point of the region it touches — for a maximum — is a vertex. So the exam algorithm is mechanical: find every corner, evaluate Z at each, pick the winner. Interior points can always be improved by sliding toward an edge, which is why vertices are the only candidates.

THE FIVE-STEP GRAPHICAL PIPELINE

1 Formulate $\rightarrow$ 2 Draw each boundary line $\rightarrow$ 3 Shade the common region $\rightarrow$ 4 List every corner $\rightarrow$ 5 Evaluate Z at each corner, pick the best

WORKED EXAMPLE — THE FULL METHOD, END TO END

Maximise $Z = 3x + 2y$ subject to $x + 2y \leq 10$, $3x + y \leq 15$, $x, y \geq 0$.

Solution. Lines: $x + 2y = 10$ cuts the axes at $(10, 0)$ and $(0, 5)$; $3x + y = 15$ cuts at $(5, 0)$ and $(0, 15)$. Origin test: both inequalities hold at $(0, 0)$, so shade toward the origin both times. **Corners of the feasible region:**

$$O(0,0), \quad (5,0), \quad (4,3), \quad (0,5)$$

where $(4, 3)$ is the intersection of the two constraint lines — solve them simultaneously to get it. **Evaluate:**

$$Z(0,0) = 0, \quad Z(5,0) = 15, \quad Z(4,3) = 3 \cdot 4 + 2 \cdot 3 = 18, \quad Z(0,5) = 10.$$

Maximum $Z = 18$ at $(4, 3)$. Verify $(4, 3)$: $4 + 6 = 10 \leq 10$ ✓ and $12 + 3 = 15 \leq 15$ ✓ — it sits on both boundaries, feasible and optimal. A question can ask for the **point** $(4, 3)$, the **value** 18, or both — read the stem.

Bounded, Unbounded, and Infeasible Regions

The region's shape decides what "optimal" even means, and the syllabus names the moods explicitly. **Bounded region:** a closed polygon — both a maximum and a minimum of Z exist, both sit at corners, evaluate everything. **Unbounded region:** the region runs off to infinity; a maximum may or may not exist. To **prove** a maximum exists at a corner C, slide the iso-profit line beyond C and check that the region offers nothing better — formally, some $ax + by \leq Z(C)$ holds over the whole region. Without that check, "unbounded" means the answer "Z can grow without

limit" is live. **Empty region:** the constraints contradict each other (say, $x + y \leq 2$ and $x + y \geq 5$) — no feasible solution exists at all, and that is the answer.

Minimisation problems (diet plans, cost cutting) typically live on **unbounded** regions: costs push you toward the origin, and the minimum sits at a corner where two constraint lines meet. The method is identical — corners, evaluate, pick — just aim the other way.

WORKED EXAMPLE — A MINIMISATION ON AN UNBOUNDED REGION

Minimise $Z = 200x + 400y$ subject to $x + 2y \geq 50$, $3x + 2y \geq 90$, $x, y \geq 0$.

Solution. Lines: $x + 2y = 50$ through $(50, 0)$, $(0, 25)$; $3x + 2y = 90$ through $(30, 0)$, $(0, 45)$. Both hold away from the origin — the region is unbounded upward. Corners: $(30, 0)$, the intersection of the two lines, and $(0, 45)$. Intersection: subtract $x + 2y = 50$ from $3x + 2y = 90$ to get $2x = 40$, so $x = 20$, $y = 15$. Evaluate:

$$Z(30, 0) = 6000, \quad Z(20, 15) = 4000 + 6000 = 10000, \quad Z(0, 45) = 18000.$$

Minimum $Z = 6000$ at $(30, 0)$. The origin is infeasible here — requirements say "at least", pushing the region away from $(0, 0)$. That flip, from $\leq$ to $\geq$, is the whole difference between the maximisation and minimisation sketches.

EVERYDAY EXAMPLE — FENCES AND SLIDING RULERS

Think of the feasible region as a paddock fenced by the constraint lines: the fences are straight, and non-negativity is the corner post at the origin. The objective function is a long straight **ruler** you slide across the paddock, always parallel to itself; where it last touches the fence line as you push it outward is the best spot in the paddock — and a fence meeting point, never the open middle. If the paddock has no far side (unbounded), the ruler can run away forever unless a fence stops it — which is exactly the existence check. Corners decide; the ruler just finds which corner.

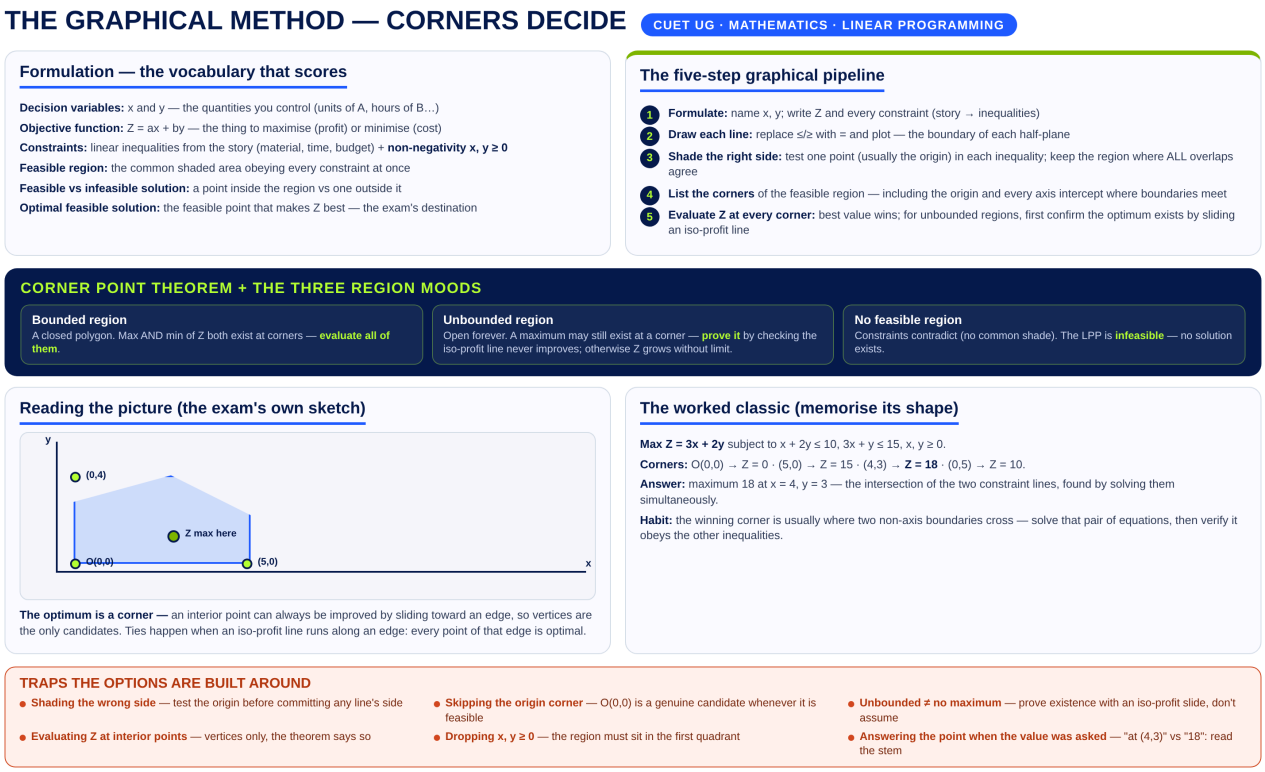


Figure 1: The graphical method on one board: formulation vocabulary, the five-step pipeline, the three region moods, and the classic worked sketch.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — LINEAR PROGRAMMING

Formula / rule	Letters mean	Fire it when
"at most / within" → $\leq$; "at least / no less" → $\geq$	translation dictionary	formulation
Objective: $Z = ax + by$	per-unit profit or cost	scoreboard to optimise
Non-negativity: $x \geq 0$, $y \geq 0$	first quadrant only	every LPP, no exceptions
Test point (origin) decides the shading side	true at origin = origin's side	drawing half-planes
Feasible region = overlap of ALL half-planes	every constraint at once	region sketches
Corner point theorem: optimum at a vertex	sliding iso-profit ruler	every optimisation

Formula / rule	Letters mean	Fire it when
Corner from two lines: solve simultaneously	intersection algebra	vertex hunting
Evaluate Z at EVERY corner, then pick	no interior candidates	final selection
Bounded: max and min both exist	closed polygon	region moods
Unbounded: maximum needs existence proof	iso-profit slide check	open regions
Empty overlap → infeasible LPP	contradicting constraints	no-region stems
Iso-profit parallel to an edge → many optima	every edge point optimal	tie questions

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — LINEAR PROGRAMMING

The mistake	What is right	The trap it sets
Shading the wrong side of a line	Test the origin first, every time	Mirrored regions and wrong corners
Dropping the non-negativity constraints	$x, y \geq 0$ always applies	Regions spilling into negative territory
Forgetting the origin as a corner	O(0,0) is a candidate when feasible	Skipped-corner option lists
Evaluating Z at interior points	Vertices only — the theorem	Attractive-but-losing interior options
Declaring unbounded = no maximum	Prove it with the iso-profit check	False "no solution" answers
Answering the point when the value was asked	Read the stem: point, value, or both	(4,3) vs 18 confusion
Missing the intersection corner	Solve the two boundary lines together	Three-of-four corner lists

The mistake	What is right	The trap it sets
Treating ties as errors	Parallel iso-profit = entire edge optimal	"Unique optimum" assumptions

Sixty-Second Self-Test

Statement	Verdict
The optimum of Z can occur at an interior point of the region.	False — vertices only
$x + y \leq 4$ and $x + y \geq 5$ give a feasible region.	False — empty region, infeasible LPP
A minimisation problem's region is often unbounded.	True — "at least" pushes away from the origin
The origin always lies in the feasible region.	False — "at least" constraints exclude it
Z must be evaluated at every listed corner.	True — the winner can be any of them
A tie means the method failed.	False — an edge of optima is a valid answer

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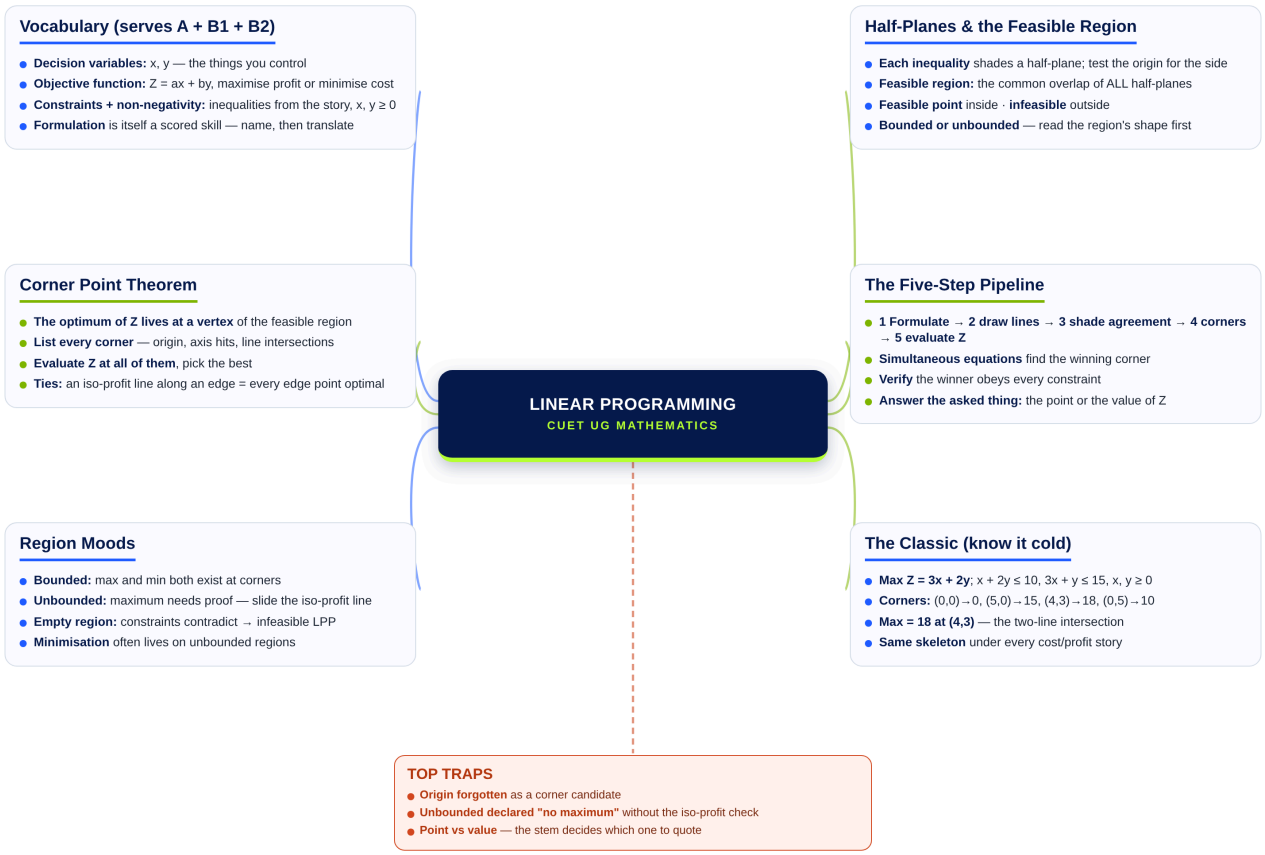


Figure 2: The chapter in one look: vocabulary, half-planes, the corner theorem, the pipeline and the classic — with traps under each.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Sketch tiny — even a 20-second corner list settles most of these.

Q1. Maximise $Z = 3x + 2y$ subject to $x + 2y \leq 10$, $3x + y \leq 15$, $x, y \geq 0$. The maximum value of Z is:

(A) 15 (B) 18
(C) 10 (D) 21

Q2. By the corner point theorem, an optimal value of the objective function (when it exists) occurs:

(A) at the centre of the feasible region (B) at the origin only
(C) on the boundary line of the first constraint (D) at a vertex of the feasible region

Q3. For the objective function $Z = 5x + 3y$, the value of Z at the origin is:

(A) 0 **(B)** 3
(C) 5 **(D)** 8

Q4. Which point is feasible for the constraints $x + y \le 4$, $2x + y \le 6$, $x, y \ge 0$?

(A) $(3, 2)$ **(B)** $(2, 3)$
(C) $(1, 1)$ **(D)** $(0, 5)$

Q5. A minimisation LPP (diet-type problem with "at least" requirements) typically has a feasible region that is:

(A) bounded **(B)** unbounded
(C) empty **(D)** a single point

Q6. For the region bounded by $x + 2y \le 10$, $3x + y \le 15$ and $x, y \ge 0$, the number of corner points is:

(A) 3 **(B)** 5
(C) 4 **(D)** 2

Q7. An LPP has infinitely many optimal solutions when:

(A) the region is unbounded **(B)** the feasible region is empty
(C) the iso-profit line is parallel to an edge of the feasible region **(D)** the origin is infeasible

Q8. For $Z = 5x + 4y$, the value of Z at the point $(2, 3)$ is:

(A) 22 **(B)** 10
(C) 12 **(D)** 23

Q9. If the constraint half-planes have no common overlap, the LPP is:

(A) unbounded **(B)** degenerate
(C) infeasible **(D)** solved by the origin

Q10. Maximise $Z = x + y$ subject to $x + 2y \le 4$, $x + y \le 3$, $x, y \ge 0$. The maximum value of Z is:

(A) 2 **(B)** 3
(C) 4 **(D)** 7

Detailed Solutions

Sol. Q1. Corners: $(0, 0) \to 0$, $(5, 0) \to 15$, $(4, 3) \to 18$, $(0, 5) \to 10$, where $(4, 3)$ solves the two constraint lines simultaneously. Maximum $Z = 18$. The interior and other corners all trail.

Answer: (B)

Sol. Q2. The sliding iso-profit ruler always leaves the region last at a vertex — interior points can be improved by sliding toward an edge. That is the theorem in one sentence.

Answer: (D)

Sol. Q3. Substitute $x = 0$, $y = 0$: $Z = 5 \cdot 0 + 3 \cdot 0 = 0$. The origin contributes nothing to either product — free marks when it is feasible.

Answer: (A)

Sol. Q4. Test each: $(3, 2)$: $5 > 4$ fails the first. $(2, 3)$: $5 > 4$ fails. $(0, 5)$: $5 > 4$ fails. $(1, 1)$: $1 + 1 = 2 \leq 4$ ✓ and $2 + 1 = 3 \leq 6$ ✓ — feasible. One substitution per option, twelve seconds total.

Answer: (C)

Sol. Q5. "At least" requirements push the region away from the origin, and it runs off outward — unbounded. The minimum still exists at a corner where two requirement lines meet.

Answer: (B)

Sol. Q6. Corners: $(0, 0)$, $(5, 0)$, the intersection $(4, 3)$, and $(0, 5)$ — four. Counting the axis intercepts and the line intersection covers every vertex of this two-constraint region.

Answer: (C)

Sol. Q7. If the iso-profit line's slope equals an edge's slope, sliding it exits along that whole edge — every point of the edge gives the same optimal Z. Geometric cause, infinite optima.

Answer: (A)

Sol. Q8. Substitute: $Z = 5 \cdot 2 + 4 \cdot 3 = 10 + 12 = 22$. Pure evaluation — the marks are in the arithmetic care.

Answer: (D)

Sol. Q9. No common overlap means no point obeys every constraint — no feasible solution exists, the LPP is infeasible. "Unbounded" would mean the region exists but runs forever.

Answer: (C)

Sol. Q10. Corners: $(0, 0) \to 0$, $(3, 0) \to 3$, $(2, 1) \to 3$, $(0, 2) \to 2$. The maximum value is 3, attained at BOTH $(3, 0)$ and $(2, 1)$ — the iso-profit line $x + y = 3$ runs along the region's edge. The question wants the value: 3.

Answer: (B)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q1	Stopped at the axis corner $(5, 0)$	The two-line intersection often wins — always find it
(A) in Q3	Added the coefficients instead of substituting	Substitute the point; Z is a machine, not a sum of knobs
(A) or (B) in Q4	Checked one constraint only	Feasible = ALL constraints — test each, reject on first failure
(A) in Q5	Pattern-matched "LPP = closed polygon"	Read the inequality directions: "at least" pushes the region out
(D) in Q6	Counted only the axis intercepts	Vertex inventory = origin + axis hits + intersections
(C) in Q10	Assumed the max sits on an axis	Ties exist; the value repeats along the whole optimal edge

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